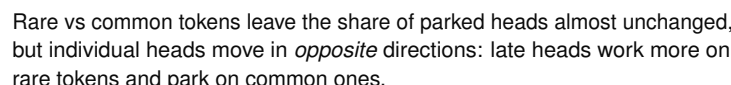


From a random-token “no-op” baseline to a phrase-sensitive effect, and the circuit test that comes next

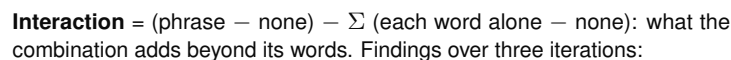
The question. At the final token of a sequence, some attention heads do work and others park their attention on the first token (BOS, the “sink”), which acts as a *no-op*. **What switches a head into the no-op state?** We feed GPT-2 small a BOS token plus 32 random tokens and measure, for each of the 144 heads, the attention the final token pays to BOS (*sink mass*). Then we change the input in controlled ways and watch which heads park more or less.

On rare-token noise (ids 1000–39999), **74% of heads park on BOS** at the final token, and parking grows with depth:



- The late-head shift comes from the **whole context**, not from the final token alone.
- Swapping one context token barely matters, **except right before the final token**: single-token effects there are about $6\times$ larger than mid-sequence (median per-head spread .029 vs .005).

Each base row is overwritten at positions 29–31; everything else, including the final token, stays the same, so every effect is a within-row difference. A real example row (rare base):



- Near the final token, a phrase makes **layer 6–11 heads park more**; written mid-sequence (positions 14–16) it does little.
- The phrase adds sink **beyond its single words**; the scrambled order adds much less. Single-word effects explain none of the order effect ($r^2 \approx 0$).
- **Not surprisal**: the model's surprise at the phrase explains about a tenth of the order effect, and across phrases the effect does not track it.

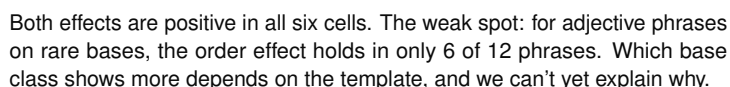
Working hypothesis. Late heads park on BOS (go no-op) when the words right before the final token form a **coherent adjacent pair**. Open question: is that no-op *computed* by a circuit, or a passive fallback?

design and predictions: exp2/testb.md

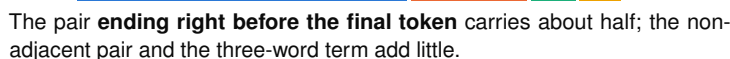
Compare “verb + object” (like `sushi`) with “object alone”: the runs differ only at position 30. Patch activations from one run into the other:

1. residual stream by position $\times$ layer: where does the signal live?
2. query vs key of the late heads: does the final token ask differently, or do the phrase tokens look different?
3. single early-head outputs: who writes it?
4. switch those heads off: does the extra parking disappear?

Late-layer interaction, mean over 12 phrases per template, 95% CI over phrases:



The phrase's interaction splits exactly into word pairs plus a three-word term (share of the total):



- **35 of 43 pre-registered predictions held.** The misses were real surprises: word order mattered more than predicted, mid-sequence placement was not a full no-op, verb-object carried 46% (predicted $\geq 50\%$), the adjective/rare cell above, plus two hairline sanity misses.
- TransformerLens and Hugging Face agree to $< 10^{-4}$ on every run. The no-phrase rows reproduce Experiment 1 ($r \geq 0.9993$), re-runs are bit-identical, and the random-word placebo flags at most 0.3% of tests in any run.
- Limits: GPT-2 small only, 3 hand-picked templates, and small effects (+0.01 to +0.03 mean sink mass over the late heads).

Pre-registered predictions

- B1 The signal moves from position 30 to 31 in early layers (31 recovers $\geq 50\%$ from layer 4 on).
- B2 Key side: patching the late heads' keys at 30–31 recovers more than their query at 32.
- B3 The top writers include a known previous-token head (e.g. L4H11, L2H2).
- B4 Switching off the top 3 writers removes $\geq 50\%$ of the effect; 3 random heads $< 20\%$.

Why it matters. It decides whether the no-op is computed or a fallback, whether heads “find nothing to read” or “stop asking”, and it gives Experiment 3 heads to switch off or patch in.

